



# Backend Mastery — Week-by-Week Calendar

**Start date:** Monday, June 30, 2026 **Method:** Follow the [5-step daily method] (file:///C:/Users/DELL/.gemini/antigravity-ide/brain/40fd8cae-88d7-4d2a-959f-7ba5d9b2b800/backend\_learning\_path.md) exactly **Two guides:**

- **Order** → [backend\_learning\_path.md](file:///C:/Users/DELL/.gemini/antigravity-ide/brain/40fd8cae-88d7-4d2a-959f-7ba5d9b2b800/backend\_learning\_path.md)
- **Depth** → [backend\_mastery\_guide.md](file:///C:/Users/DELL/.gemini/antigravity-ide/brain/40fd8cae-88d7-4d2a-959f-7ba5d9b2b800/backend\_mastery\_guide.md)

[!TIP] **You already built a Finance Tracker.** You probably know Steps 1–8 in practice. Still go through them — but focus on **gaps** (recursion depth, Big-O, decorators). You can do 2 steps/day for familiar topics. This could compress Milestone 1 from 6 weeks to 2 weeks, finishing the full path in ~8 months instead of 12.

## How to Read This Calendar

| Symbol                                                                              | Meaning                            |
|-------------------------------------------------------------------------------------|------------------------------------|
|  | Study day — read + notes           |
|  | Build day — code implementation    |
|  | Revision day — fix weak spots      |
|  | Checkpoint — test yourself         |
|  | Project day — build something real |

Each day has:

- **What to do** (specific step + focus area)
- **Output** (what you should produce)
- **Time** (~2-3 hours)

## MILESTONE 1: Programming Foundations

### Week 1 — Variables, Control Flow, Functions

| Day | Type                                                                                                                                                                       | Step          | Focus                                         | Output                                                  |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----------------------------------------------|---------------------------------------------------------|
| Mon | <br> | <b>Step 1</b> | Variables, types, operators, strings          | 5 notes + script: calculator that handles +, -, *, /, % |
| Tue | <br> | <b>Step 2</b> | if/elif/else, for/while loops, break/continue | 5 notes + script: FizzBuzz + number guessing game       |

| Day | Type                                                                                                                                                                   | Step                   | Focus                                         | Output                                                                            |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------------------------------|-----------------------------------------------------------------------------------|
| Wed | <br> | <b>Step 3 (part 1)</b> | Functions, params, return, scope, docstrings  | 5 notes + script: 5 utility functions (is_prime, factorial, reverse_string, etc.) |
| Thu | <br> | <b>Step 3 (part 2)</b> | Lambda, decorators, recursion, *args/**kwargs | 5 notes + script: decorator that logs function calls + recursive fibonacci        |
| Fri | <br> | <b>Step 4 (part 1)</b> | Lists, list comprehensions, sorting, slicing  | 5 notes + script: list operations (filter, transform, search)                     |
| Sat |                                                                                       | Review Steps 1–4       | Fix anything you couldn't explain clearly     | Re-do any weak exercise                                                           |
| Sun |                                                                                       | <b>Step 4 (part 2)</b> | Dicts, sets, tuples, when-to-use-each         | Script: word frequency counter using dicts + sets                                 |

## Week 2 — OOP, Error Handling, Modules

| Day | Type                                                                                                                                                                       | Step                   | Focus                                                    | Output                                                                                                   |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Mon | <br>    | <b>Step 5 (part 1)</b> | Classes, <b>init</b> , self, methods, properties         | 5 notes + class: <b>BankAccount</b> with deposit, withdraw, balance                                      |
| Tue | <br> | <b>Step 5 (part 2)</b> | Inheritance, composition, polymorphism, abstract classes | 5 notes + class hierarchy: <b>Shape</b> → <b>Circle</b> , <b>Rectangle</b> with area()                   |
| Wed | <br> | <b>Step 6</b>          | try/except, custom exceptions, context managers          | 5 notes + script: file processor that handles missing files, bad formats, custom <b>InvalidDataError</b> |
| Thu | <br> | <b>Step 7 (part 1)</b> | Modules, packages, virtual environments, pip             | 5 notes + create a Python package with <b>__init__.py</b> , install external package                     |
| Fri | <br> | <b>Step 7 (part 2)</b> | File I/O (CSV, JSON), env vars, datetime                 | 5 notes + script: read CSV of expenses, parse dates, write JSON summary                                  |
| Sat |                                                                                         | Review Steps 5–7       | Can you explain OOP inheritance to someone?              | Re-do any weak exercise                                                                                  |
| Sun | <br> | <b>Step 8</b>          | Big-O, binary search, hash tables                        | 5 notes + implement binary search + explain why dict lookup is O(1)                                      |

## Week 3 — MILESTONE 1 CHECKPOINT

| Day | Type                                                                                | Step                    | Focus                                                             | Output                                           |
|-----|-------------------------------------------------------------------------------------|-------------------------|-------------------------------------------------------------------|--------------------------------------------------|
| Mon |  | <b>Checkpoint build</b> | Build CLI expense tracker (add, list, delete, filter by category) | Working script using all concepts from Steps 1–8 |

| Day | Type                                                                                                                                                                   | Step                    | Focus                                                    | Output                                                                   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------|--------------------------------------------------------------------------|
| Tue |                                                                                       | <b>Checkpoint build</b> | Add: save to CSV, load from CSV, monthly summary         | Complete CLI tool                                                        |
| Wed |                                                                                       | <b>Self-test</b>        | Explain: OOP, scope, Big-O, dict vs list, error handling | Write 1-page summary of what you learned                                 |
| Thu | <br> | <b>Step 9</b>           | Internet basics — IP, ports, DNS, TCP handshake          | 5 notes + use <code>nslookup</code> , <code>ping</code> , identify ports |
| Fri | <br> | <b>Step 10 (part 1)</b> | HTTP request/response structure, methods, status codes   | 5 notes + make 5 <code>curl</code> requests to a public API              |
| Sat |                                                                                       | Review                  | Can you explain TCP 3-way handshake? HTTP methods?       | Fix any gaps                                                             |
| Sun | <br> | <b>Step 10 (part 2)</b> | Headers, JSON, cookies, content negotiation              | 5 notes + curl with custom headers, parse JSON response                  |

## MILESTONE 2: How the Web Works

### Week 4 — Web Fundamentals Complete

| Day | Type                                                                                                                                                                       | Step                          | Focus                                                             | Output                                                                    |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| Mon | <br> | <b>Step 11</b>                | HTTPS/TLS, cookies (HttpOnly, Secure, SameSite), sessions         | 5 notes + diagram: "How login works with sessions"                        |
| Tue |                                                                                         | <b>Milestone 2 checkpoint</b> | Explain: "What happens when you type google.com?"                 | Write full answer (DNS → TCP → TLS → HTTP → response)                     |
| Wed | <br> | <b>Step 12 (part 1)</b>       | Django setup, project structure, urls.py, views.py                | 5 notes + running Django project with one "hello world" view              |
| Thu | <br> | <b>Step 12 (part 2)</b>       | Request-response cycle, templates, static files                   | 5 notes + page that displays current time from a view                     |
| Fri | <br> | <b>Step 13 (part 1)</b>       | Database concepts, PostgreSQL setup, CREATE TABLE, INSERT, SELECT | 5 notes + create a <code>transactions</code> table in psql, insert 5 rows |
| Sat |                                                                                         | Review Steps 9–13             | Can you trace a full HTTP request through Django?                 | Fix any gaps                                                              |
| Sun | <br> | <b>Step 13 (part 2)</b>       | WHERE, ORDER BY, GROUP BY, HAVING, aggregates                     | 5 notes + write 10 SQL queries against your transactions table            |

## MILESTONE 3: Your First Backend

## Week 5 — ORM & REST Basics

| Day | Type                                                                                                                                                                   | Step                    | Focus                                                | Output                                                                    |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------------------------------------|---------------------------------------------------------------------------|
| Mon | <br> | <b>Step 14 (part 1)</b> | Django models, field types, migrations               | 5 notes + define 3 models (Category, Transaction, Budget), migrate        |
| Tue | <br> | <b>Step 14 (part 2)</b> | ForeignKey, queries, filter, select_related, N+1     | 5 notes + Django shell: create related objects, query with select_related |
| Wed | <br> | <b>Step 15 (part 1)</b> | REST principles, CRUD endpoints (GET, POST)          | 5 notes + build list + create views for transactions                      |
| Thu | <br> | <b>Step 15 (part 2)</b> | PUT/PATCH/DELETE, status codes, error responses      | 5 notes + complete CRUD for transactions                                  |
| Fri | <br> | <b>Step 16</b>          | Layered architecture: thin views → services → models | 5 notes + refactor: move business logic from views to services.py         |
| Sat |                                                                                       | Review Steps 14–16      | Can you explain: ORM vs raw SQL? Why thin views?     | Fix any gaps                                                              |
| Sun |                                                                                     | <b>Practice</b>         | Build a mini CRUD API for a new entity (e.g., Notes) | Working CRUD with service layer                                           |

## Week 6 — SQL Deep Dive + Milestone 3 Checkpoint

| Day | Type                                                                                                                                                                       | Step                          | Focus                                                          | Output                                                                 |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------|------------------------------------------------------------------------|
| Mon | <br> | <b>Step 17 (part 1)</b>       | JOINS (INNER, LEFT), subqueries, CTEs                          | 5 notes + write 5 JOIN queries combining transactions + categories     |
| Tue | <br> | <b>Step 17 (part 2)</b>       | Indexes, EXPLAIN ANALYZE, ACID transactions                    | 5 notes + add indexes, run EXPLAIN on slow queries, see the difference |
| Wed |                                                                                         | <b>Milestone 3 checkpoint</b> | Can you build CRUD from scratch? Explain indexes? Write JOINS? | Self-test: build a new mini API in 2 hours                             |
| Thu | <br> | <b>Step 18 (part 1)</b>       | Registration, password hashing, login flow                     | 5 notes + build registration + login views                             |
| Fri | <br> | <b>Step 18 (part 2)</b>       | Session auth, JWT concepts, @login_required, logout            | 5 notes + protect views, test logged-in vs logged-out                  |
| Sat |                                                                                         | Review Steps 17–18            | Can you explain: isolation levels? Session vs JWT?             | Fix any gaps                                                           |
| Sun |                                                                                         | <b>Practice</b>               | Add auth to your Notes API from last Sunday                    | Working auth flow                                                      |

## MILESTONE 4: Make It Real

### Week 7 — Authorization, Validation, Config

| Day | Type                                                                                                                                                                   | Step                          | Focus                                                                | Output                                                                 |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------|
| Mon |      | <b>Step 19</b>                | Authorization, RBAC, object-level permissions, data isolation        | 5 notes + add ownership checks: users only see their own data          |
| Tue |      | <b>Step 20</b>                | Input validation: form + model + DB constraints                      | 5 notes + add validation to Transaction (no zero, no negative income)  |
| Wed |      | <b>Step 21</b>                | .env files, python-decouple, 12-Factor App basics                    | 5 notes + move all secrets to .env, create .env.example                |
| Thu |      | <b>Step 22 (part 1)</b>       | Notification system, email sending (Resend/SendGrid)                 | 5 notes + create Notification model, send test email                   |
| Fri |   | <b>Step 22 (part 2)</b>       | File uploads, external API integration, fallback pattern             | 5 notes + add file upload + graceful API call with try/except fallback |
| Sat |                                                                                     | Review Steps 19–22            | Can you explain: auth vs authz? Validation layers? Fallback pattern? | Fix any gaps                                                           |
| Sun |                                                                                     | <b>Milestone 4 checkpoint</b> | Build: authenticated API with validation + email notification        | Self-test with your Finance Tracker as reference                       |

## MILESTONE 5: Make It Safe & Tested

### Week 8 — Security, Testing, Git

| Day | Type                                                                                                                                                                    | Step                    | Focus                                                    | Output                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------|----------------------------------------------------------------|
| Mon |   | <b>Step 23</b>          | SQL injection, XSS, CSRF, security headers               | 5 notes + verify Django's protections, add security middleware |
| Tue |   | <b>Step 24 (part 1)</b> | Unit tests: test one function, AAA pattern, assertions   | 5 notes + write 5 unit tests for your service functions        |
| Wed |   | <b>Step 24 (part 2)</b> | Integration tests, view tests, mocking external services | 5 notes + write 5 integration tests, mock email sending        |
| Thu |   | <b>Step 25</b>          | Git: branch, merge, rebase, PR, .gitignore, conflicts    | 5 notes + create feature branch, make PR, resolve a conflict   |

| Day | Type                                                                                                                                                                | Step                    | Focus                                               | Output                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------|-----------------------------------------------------|
| Fri |   | <b>Step 26 (part 1)</b> | Docker: Dockerfile, build, run, .dockerignore       | 5 notes + Dockerize your app, run it in a container |
| Sat |                                                                                    | Review Steps 23–26      | Can you explain: CSRF? Test pyramid? Docker layers? | Fix any gaps                                        |
| Sun |   | <b>Step 26 (part 2)</b> | Docker Compose: multi-container (app + PostgreSQL)  | Working docker-compose.yml                          |

## Week 9 — Deployment + Milestone 5 Checkpoint

| Day | Type                                                                                                                                                                | Step                          | Focus                                                    | Output                                              |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------|-----------------------------------------------------|
| Mon |   | <b>Step 27 (part 1)</b>       | Deployment concepts, Render/Railway setup                | 5 notes + deploy your app to Render                 |
| Tue |   | <b>Step 27 (part 2)</b>       | CI/CD: GitHub Actions, production checklist              | 5 notes + add GitHub Action that runs tests on push |
| Wed |                                                                                    | <b>Milestone 5 checkpoint</b> | Your app is deployed, tested, secured, with CI/CD        | Verify: push code → tests run → app deploys         |
| Thu |                                                                                  | <b>Big review</b>             | Review Milestones 1–5 — identify weakest 3 topics        | Study notes for weak topics                         |
| Fri |                                                                                  | <b>Deep review</b>            | Re-study the 3 weakest topics                            | Re-do exercises for those topics                    |
| Sat |                                                                                  | Review                        | Final cleanup of any gaps                                | Clean notes                                         |
| Sun |                                                                                  | <b>Project</b>                | Start a fresh mini-project (Blog API) to test all skills | Skeleton with auth + CRUD + tests + Docker          |

[!IMPORTANT] 🚀 **After Week 9, you are an intermediate backend engineer.** Everything after this builds senior-level depth.

## MILESTONE 6: Make It Fast & Scalable

### Week 10 — Caching & Background Jobs

| Day | Type                                                                                                                                                                    | Step                    | Focus                                                   | Output                                                |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------|-------------------------------------------------------|
| Mon |   | <b>Step 28 (part 1)</b> | Why cache, Redis setup, cache-aside pattern             | 5 notes + install Redis, cache a slow query result    |
| Tue |   | <b>Step 28 (part 2)</b> | TTL, invalidation, cache stampede, what to/not to cache | 5 notes + implement cache invalidation on data change |

| Day | Type                                                                                                                                                                   | Step                    | Focus                                                                     | Output                                               |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------------------------|------------------------------------------------------|
| Wed | <br> | <b>Step 29 (part 1)</b> | Celery setup, define task, run worker                                     | 5 notes + create Celery task, execute it async       |
| Thu | <br> | <b>Step 29 (part 2)</b> | Retry policies, dead letter queue, scheduled tasks                        | 5 notes + add retry to task, configure periodic task |
| Fri | <br> | <b>Step 30 (part 1)</b> | Pub/sub vs queue, delivery guarantees, idempotent consumers               | 5 notes + diagram: pub/sub vs queue with examples    |
| Sat |                                                                                       | Review Steps 28–30      | Can you explain: cache-aside? Exponential backoff? Exactly-once illusion? | Fix any gaps                                         |
| Sun | <br> | <b>Step 30 (part 2)</b> | Outbox pattern, inbox pattern, Kafka basics (concepts)                    | 5 notes + implement simple dedup table for messages  |

## Week 11 — Database Performance & Monitoring

| Day | Type                                                                                                                                                                       | Step                          | Focus                                                               | Output                                                             |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|---------------------------------------------------------------------|--------------------------------------------------------------------|
| Mon | <br>  | <b>Step 31 (part 1)</b>       | N+1 problem, select_related, EXPLAIN ANALYZE                        | 5 notes + find and fix an N+1 query in your project                |
| Tue | <br> | <b>Step 31 (part 2)</b>       | Window functions, cursor pagination, connection pooling             | 5 notes + write window function query, implement cursor pagination |
| Wed | <br> | <b>Step 32 (part 1)</b>       | Structured logging, log levels, request IDs, PII redaction          | 5 notes + set up structured JSON logging in your app               |
| Thu | <br> | <b>Step 32 (part 2)</b>       | Metrics, alerts, Sentry/Prometheus basics                           | 5 notes + integrate Sentry for error tracking                      |
| Fri | <br> | <b>Step 33</b>                | NoSQL use cases, Redis data types, SQL vs NoSQL decision matrix     | 5 notes + Redis exercise: implement rate limiter with sorted sets  |
| Sat |                                                                                         | Review Steps 31–33            | Can you explain: N+1 fix? When Redis vs Mongo vs Postgres?          | Fix any gaps                                                       |
| Sun |                                                                                         | <b>Milestone 6 checkpoint</b> | Self-test: optimize a slow API (caching, query fix, background job) | Document: "3 performance improvements I made and why"              |

## MILESTONE 7: Design & Lead

## Week 12 — Scaling & Distributed Systems

| Day | Type                                                                                                                                                                | Step                    | Focus                                                                    | Output                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|
| Mon |   | <b>Step 34 (part 1)</b> | Vertical vs horizontal scaling, stateless services                       | 5 notes + diagram: "How to scale my Finance Tracker to 100K users" |
| Tue |   | <b>Step 34 (part 2)</b> | Load balancing algorithms, DB read replicas, sharding                    | 5 notes + explain: round-robin vs consistent hashing               |
| Wed |   | <b>Step 35 (part 1)</b> | CAP theorem, eventual consistency, real examples                         | 5 notes + classify 5 databases as CP or AP with reasoning          |
| Thu |   | <b>Step 35 (part 2)</b> | Distributed locking, idempotency, consensus (Raft), clock problems       | 5 notes + explain: why Redis lock needs fencing tokens             |
| Fri |   | <b>Step 36</b>          | Timeouts, retries, circuit breaker, bulkhead, graceful degradation       | 5 notes + diagram: circuit breaker state machine                   |
| Sat |                                                                                    | Review Steps 34–36      | Can you explain: CAP? Circuit breaker? Why clocks lie?                   | Fix any gaps                                                       |
| Sun |   | <b>Step 37</b>          | Architecture patterns: monolith, microservices, CQRS, saga, outbox+inbox | 5 notes + compare: when monolith vs microservices                  |

## Week 13 — Cloud, Kubernetes & System Design Prep

| Day | Type                                                                                                                                                                    | Step                    | Focus                                                                | Output                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------------------|
| Mon |   | <b>Step 38 (part 1)</b> | Kubernetes basics: pods, deployments, services, health probes        | 5 notes + deploy a container to a local k8s (minikube) or read k8s YAML configs |
| Tue |   | <b>Step 38 (part 2)</b> | Cloud services (AWS/GCP overview), IaC basics, deployment strategies | 5 notes + diagram: "My app on AWS" (EC2, RDS, ElastiCache, S3)                  |
| Wed |   | <b>Step 39 (part 1)</b> | System design methodology (7 steps), capacity estimation formulas    | 5 notes + practice: estimate capacity for 1M-user app                           |
| Thu |                                                                                      | <b>Step 39 (part 2)</b> | Design practice: URL shortener                                       | Full system design write-up (requirements, API, data model, components)         |
| Fri |                                                                                      | <b>Step 39 (part 3)</b> | Design practice: notification system                                 | Full system design write-up                                                     |
| Sat |                                                                                      | Review Steps 37–39      | Practice explaining your URL shortener design out loud               | Fix any gaps                                                                    |

| Day | Type | Step                   | Focus                                             | Output       |
|-----|------|------------------------|---------------------------------------------------|--------------|
| Sun | ✓    | Milestone 7 checkpoint | Self-test: design a system in 30 minutes on paper | Celebrate! 🎉 |

Weeks 14–16 — System Design Deep Practice (Optional but Recommended)

| Week | Focus                                                         | Daily Practice                            |
|------|---------------------------------------------------------------|-------------------------------------------|
| 14   | Design 3 more systems: chat app, rate limiter, payment system | One design per 2 days + review            |
| 15   | LeetCode medium problems (arrays, strings, hash maps, trees)  | 2 problems/day                            |
| 16   | Mock interviews: explain your Finance Tracker architecture    | Practice with a friend or record yourself |

Calendar Summary

| Week  | Dates           | Milestone               | Steps             | Level After                 |
|-------|-----------------|-------------------------|-------------------|-----------------------------|
| 1     | Jun 30 – Jul 6  | 1: Foundations          | 1–4               | "I can write Python"        |
| 2     | Jul 7 – Jul 13  | 1: Foundations          | 5–8               | "I can solve problems"      |
| 3     | Jul 14 – Jul 20 | 1→2: Checkpoint + Web   | Checkpoint + 9–10 | "I understand the web"      |
| 4     | Jul 21 – Jul 27 | 2→3: Web + Backend      | 11–13             | "I know HTTP + SQL"         |
| 5     | Jul 28 – Aug 3  | 3: First Backend        | 14–16             | "I can build APIs"          |
| 6     | Aug 4 – Aug 10  | 3→4: SQL + Auth         | 17–18             | "I can write JOINS + login" |
| 7     | Aug 11 – Aug 17 | 4: Real Features        | 19–22             | "I can build real apps"     |
| 8     | Aug 18 – Aug 24 | 5: Safe & Tested        | 23–26             | "I can test + Dockerize"    |
| 9     | Aug 25 – Aug 31 | 5: Deploy               | 27 + review       | INTERMEDIATE ✓              |
| 10    | Sep 1 – Sep 7   | 6: Performance          | 28–30             | "I know caching + async"    |
| 11    | Sep 8 – Sep 14  | 6: DB perf + monitoring | 31–33             | "I can optimize systems"    |
| 12    | Sep 15 – Sep 21 | 7: Distributed + Design | 34–37             | "I understand scale"        |
| 13    | Sep 22 – Sep 28 | 7: Cloud + Design       | 38–39             | SENIOR-READY ✓              |
| 14–16 | Sep 29 – Oct 19 | Bonus: Interview prep   | Practice          | INTERVIEW-READY 🎯           |

## Fast-Track Option

Since you already built the [Finance Tracker](file:///d:/Document/Projects/FJ-BE-R2-Satyam-Keshari-IIIT-Pune), you likely know Steps 1–8 in practice. Here's an accelerated option:

| Approach                                              | Start Point | Duration |
|-------------------------------------------------------|-------------|----------|
| <b>Full path</b> (recommended for deep understanding) | Step 1      | 13 weeks |
| <b>Fast-track</b> (if Steps 1–8 feel easy)            | Step 9      | 9 weeks  |
| <b>Speed run</b> (if Steps 1–17 feel easy)            | Step 18     | 6 weeks  |

[!IMPORTANT] **Honest self-test before skipping:** For each step you want to skip, can you (1) explain it to someone, (2) write working code, and (3) answer an interview question about it? If yes → skip. If no → study it.

## Daily Template

Copy this for each day:

```
## Day: [date]
### Step: [number] – [title]

**5 Bullet Notes:**
1.
2.
3.
4.
5.

**Code Snippet:**
[save file to a `learning/` folder in your project]

**Can I explain this simply?** [ ] Yes [ ] Not yet
**Can I write working code?** [ ] Yes [ ] Not yet
**Checkbox marked in learning_path.md?** [ ] Yes
```

[!TIP] **Start tomorrow (Monday, June 30).** Open [backend\_learning\_path.md] (file:///C:/Users/DELL/.gemini/antigravity-ide/brain/40fd8cae-88d7-4d2a-959f-7ba5d9b2b800/backend\_learning\_path.md), go to Step 1, follow the daily template above. That's it. One step at a time.